

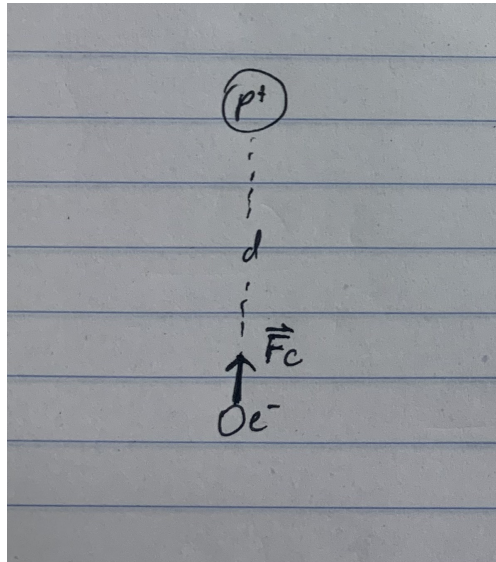
Homework 1

Andrew Hwang
Physics 16
Professor Zhou

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Question 1. Purcell chapter 1, problem 35 (Balancing the weight). On the utterly unrealistic assumption that there are no other charged particles in the vicinity, at what distance below a proton would the upward force on an electron equal the electron's weight?

Solution. We draw the problem as shown below.



Assuming that the electron's weight refers to the weight on Earth, where $g = 9.8 \frac{\text{m}}{\text{s}^2}$, we have to make the magnitude of forces equal so that

$$k \frac{q_1 q_2}{d^2} = m_e g \quad (1)$$

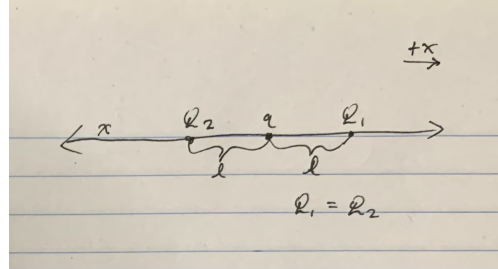
Rearranging the equation to separate d gives us

$$d = \sqrt{\frac{k q_1 q_2}{m_e g}} \approx \sqrt{\frac{(9 \times 10^9 \frac{\text{N} \cdot \text{m}^2}{\text{C}^2})(1.6 \times 10^{-19} \text{ C})^2}{(9.11 \times 10^{-31} \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2})}} \approx \boxed{5.08 \text{ m.}}$$

□

Question 2. Purcell chapter 1, problem 38 (Oscillating on a line). Two positive point charges Q are located at points $(\pm\ell, 0)$. A particle with positive charge q and mass m is initially located midway between them and is then given a tiny kick. If it is constrained to move along the line joining the two charges Q , show that it undergoes simple harmonic motion (for small oscillations), and find the frequency.

Solution. We draw the problem as such:



Let us use Newton's Second Law to describe the point charge setup. Since all charges are positive, the Coulombic forces is repulsive. Assuming the right direction is positive x as shown by the figure,

$$\begin{aligned} F = ma &= F_{Q_1q} + F_{Q_2q} = -k\frac{qQ}{(l-x)^2} + k\frac{qQ}{(l+x)^2} \\ &= kqQ \left[-(l-x)^{-2} + (l+x)^{-2} \right]. \end{aligned}$$

Now, I will use the Taylor approximation for $(l-x)^{-2}$ and $(l+x)^{-2}$ to the first order. For $(l-x)^{-2}$,

$$f(x) \approx f(0) + f'(0)(x) = (l-0)^{-2} - 2(l-0)^{-3}(-1)(x) = \frac{1}{l^2} + \frac{2x}{l^3}.$$

Similarly, for $(l+x)^{-2}$,

$$f(x) \approx f(0) + f'(0)(x) = (l+0)^{-2} - 2(l+0)^{-3}(x) = \frac{1}{l^2} - \frac{2x}{l^3}.$$

Substituting these approximations into Newton's Second Law, it is then

$$\begin{aligned} ma &\approx kqQ \left[-\left(\frac{1}{l^2} + \frac{2x}{l^3} \right) + \frac{1}{l^2} - \frac{2x}{l^3} \right] \\ &= -kqQ \left(\frac{4x}{l^3} \right) \end{aligned}$$

Hence, since $a = \ddot{x}$,

$$\boxed{\ddot{x} + \frac{4kqQx}{ml^3} = 0,}$$

which is a second-order differential equation for harmonic motion ($\ddot{x} + \omega^2 x = 0$). Therefore, the frequency ω is

$$\boxed{\omega = \sqrt{\frac{4kqQ}{ml^3}}.}$$

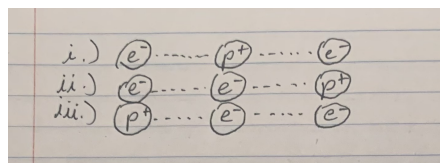
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Question 3. After Purcell chapter 1, problem 40 (Zero potential energy). Find a geometrical arrangement of one proton and two electrons such that the potential energy of the system is exactly zero. Do not do the general case, but for simplicity, only consider arrangements on a straight line. Explore all three possible permutations of arrangements of these three particles on a straight line, and calculate the positions of the three particles that lead to zero potential. You should find that the ratio of two of the distances involved is the golden ratio.

Solution. The three particles are one proton (+e) and two electrons (-e). The potential energy of a three-particle system is

$$U = k \left(\frac{q_1 q_2}{r_{12}} + \frac{q_1 q_3}{r_{13}} + \frac{q_2 q_3}{r_{23}} \right) = 0.$$

The three possible linear arrangements are:



However, because there is no loss of generality between cases (ii) and (iii), there are really two distinct cases to consider. In (i), placing the proton at the center,

$$U = k \left(\frac{(-e)(+e)}{a} + \frac{(-e)(+e)}{b} + \frac{(-e)(-e)}{a+b} \right) = \frac{ke^2}{a+b} - \frac{ke^2}{a} - \frac{ke^2}{b} = 0$$

$$ab - b(a+b) - a(a+b) = 0 \implies a^2 + ab + b^2 = 0.$$

This has no real positive solutions, so there are no possible arrangements in such case. In (ii), placing the proton in the right end,

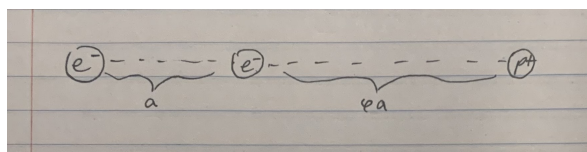
$$U = k \left(\frac{(-e)(-e)}{a} + \frac{(-e)(+e)}{a+b} + \frac{(-e)(+e)}{b} \right) = \frac{ke^2}{a} - \frac{ke^2}{a+b} - \frac{ke^2}{b} = 0$$

$$\implies \frac{1}{a} - \frac{1}{a+b} - \frac{1}{b} = 0 \implies b^2 - ab - a^2 = 0.$$

Dividing through by a^2 and letting $\phi = b/a$,

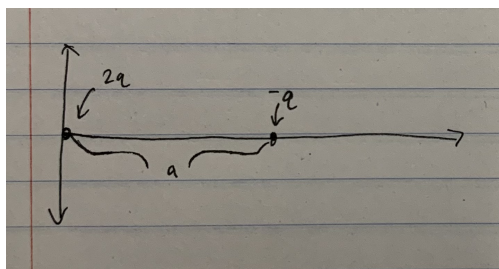
$$\phi^2 - \phi - 1 = 0 \implies \boxed{\phi = \frac{1 + \sqrt{5}}{2} \equiv \varphi \approx 1.618.}$$

Therefore, the arrangement is shown below. The mirror image (iii) gives similar results. □



Question 4. Purcell chapter 1, exercise 57 (Escaping field lines). Charges $2q$ and $-q$ are located on the x axis at $x = 0$ and $x = a$, respectively.

- Find the point on the x axis where the electric field is zero, and make a rough sketch of some field lines.
- You should find that some of the field lines that start on the $2q$ charge end up on the $-q$ charge, while others head off to infinity. Consider the field lines that form the cutoff between these two cases. At what angle (with respect to the x axis) do these lines leave the $2q$ charge? *Hint:* Draw a wisely chosen Gaussian surface that mainly follows these lines.



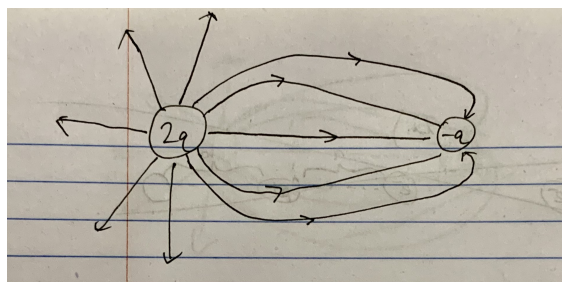
Solution. For part (a), the electric field on the x -axis due to both charges is

$$E = \frac{k(2q)}{x^2} - \frac{kq}{(x-a)^2} = 0,$$

where we look for a point x where the two contributions cancel. When putting a point charge inside, the positive charge pushes right and the negative pulls left. Therefore, the zero must lie outside the two charges. Since $|2q| > |-q|$, the zero lies at $x > a$. Setting the field to zero,

$$\begin{aligned} \frac{2q}{x^2} &= \frac{q}{(x-a)^2} \implies 2(x-a)^2 = x^2 \\ \implies \sqrt{2}(x-a) &= x \implies x = \frac{\sqrt{2}}{\sqrt{2}-1} a = (2 + \sqrt{2})a. \end{aligned}$$

The fields lines look like such:



For part (b), inside the cutoff, the field lines point towards the $-q$ point charge. To find the

angle θ , we will look at boundary conditions. We define a Gaussian surface consisting of a spherical cap centered at $x = 0$ with a half-angle θ_{critical} . The flux Φ_E through this spherical cap is

$$\begin{aligned}
 \Phi_E &= \int_0^{2\pi} \int_0^{\theta_{\text{critical}}} \frac{2q}{4\pi\epsilon_0 r^2} (r^2 \sin \theta d\theta d\phi) \\
 &= \frac{2q}{4\pi\epsilon_0} \int_0^{2\pi} d\phi \int_0^{\theta_{\text{critical}}} \sin \theta d\theta \\
 &= \frac{2q}{4\pi\epsilon_0} (2\pi) [-\cos \theta]_0^{\theta_{\text{critical}}} \\
 &\quad \boxed{\Phi_E = \frac{q}{\epsilon_0} (1 - \cos \theta_{\text{critical}}).}
 \end{aligned}$$

By Gauss's Law, the total flux of the system is

$$\Phi_E = \frac{2q + (-q)}{\epsilon_0} = \frac{q}{\epsilon_0}.$$

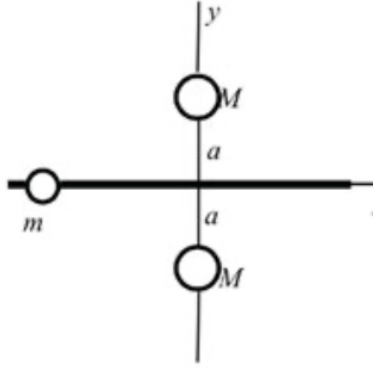
At the critical angle, all field lines within the cone that leave $2q$ must terminate on $-q$, so the flux through the cap equals the total flux. Therefore,

$$\frac{q}{\epsilon_0} (1 - \cos \theta_{\text{critical}}) = \frac{q}{\epsilon_0} \implies \cos \theta_{\text{critical}} = 0 \implies \boxed{\theta_{\text{critical}} = \frac{\pi}{2}}.$$

This result makes physical sense. Sufficiently close to the $2q$ charge, the field lines appear uniformly distributed in all directions, as the influence of $-q$ is negligible at short range. Exactly half of the $2q$ field lines must terminate on $-q$. Therefore, it would make sense that the field lines in the right hemisphere (the hemisphere closer to the $-q$ charge) would terminate on $-q$, while those in the left hemisphere escape to infinity. In spherical coordinates, the right hemisphere would be represented by $(-\frac{\pi}{2} < \theta < \frac{\pi}{2})$. $\square$

Question 5. A bead of mass m and charge $-q$ slides without friction on a smooth rod along the x -axis. The rod is equidistant between two spheres of charge Q and mass M . The spheres are located at $x = 0$, $y = \pm a$ as shown, and attract the bead electrically.

- (a) Find the potential energy of the bead as a function of its position, x .
- (b) The bead is released at $x = -3a$ with velocity v_0 toward the origin. Find the speed as it passes through the origin and describe qualitatively the motion of the bead.
- (c) Find the frequency of small oscillations of the bead about the origin.



Solution. For part (a), it is true that there are two forces present: gravity and electrostatic. Recall that the gravitational potential U_G is

$$U_g = -\frac{GMm}{r},$$

where $r = \sqrt{x^2 + a^2}$ using Pythagorean's theorem. Therefore, the total potential energy U as a function of position is

$$U(x) = U_G + U_E = -\frac{2GMm}{\sqrt{x^2 + a^2}} + 2(-q)V = -\frac{2GMm}{\sqrt{x^2 + a^2}} - \frac{2kQq}{\sqrt{x^2 + a^2}}$$

$$U(x) = -\frac{2(GMm + kQq)}{\sqrt{x^2 + a^2}}.$$

For part (b), the mechanical energy of the system is conserved, thus

$$E = U + K = \frac{1}{2}mv_0^2 - \frac{2(GMm + kQq)}{\sqrt{(-3a)^2 + a^2}} = \frac{1}{2}mv_f^2 - \frac{2(GMm + kQq)}{\sqrt{(0)^2 + a^2}}.$$

Rearranging the equation gives

$$v_f^2 = v_0^2 + \frac{4(GMm + kQq)}{ma} \left(1 - \frac{1}{\sqrt{10}}\right) \Rightarrow v_f = \sqrt{v_0^2 + \frac{4(GMm + kQq)}{ma} \left(1 - \frac{1}{\sqrt{10}}\right)}.$$

For part (c), we must use the approximation of $\frac{d^2U}{dx^2} \approx k$:

$$\begin{aligned}\frac{dU}{dx} &= -2(GMm + kQq) \frac{d}{dx} (x^2 + a^2)^{-1/2} = -2(GMm + kQq) \left[-\frac{1}{2} (x^2 + a^2)^{-3/2} \right] (2x) \\ &= 2x(GMm + kQq)(x^2 + a^2)^{-3/2} \\ \frac{d^2U}{dx^2} &= 2(GMm + kQq) \frac{d}{dx} (x(x^2 + a^2)^{-3/2}) = 2(GMm + kQq) \left[-\frac{3}{2} x(x^2 + a^2)^{-5/2} + (x^2 + a^2)^{-3/2} \right]\end{aligned}$$

Hence,

$$k \approx \frac{d^2U}{dx^2}(0) = 2(GMm + kQq) [0 + (0^2 + a^2)^{-3/2}] = \frac{2(GMm + kQq)}{a^3}$$

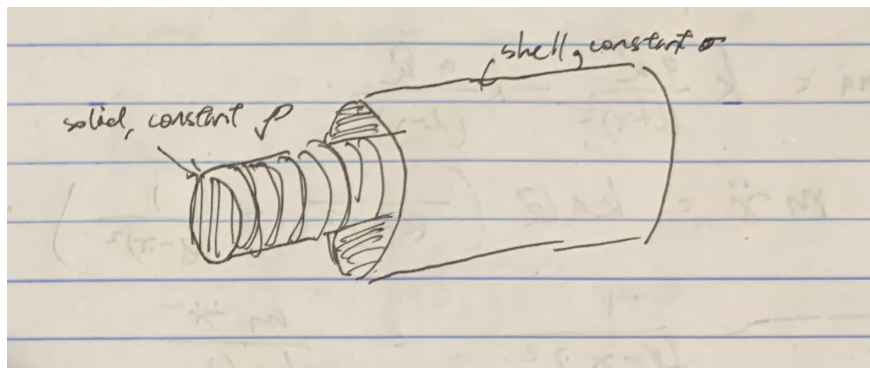
Therefore, the frequency of oscillations ω is

$$\omega = \sqrt{\frac{k}{m}} \approx \sqrt{\frac{2(GMm + kQq)}{ma^3}}.$$

□

Question 6. A coaxial cable is made of a long cylindrical copper wire surrounded by a conducting cylindrical shell. Initially, the wire carries positive volume charge density ρ and the outer shell carries negative surface charge density σ . The net charge including both cylinders is zero. Find the initial electric field in each region: inside the wire, between the center conductor and the shell, and outside the cable.

Solution. The wire is described as such: All of the electric fields can be solved using Gauss'



Law. Let us define R_1 to be the radius of the inner cable, and R_2 to be the radius of the outer shell. Additionally, let us define H to be the length of the wire and shell. Firstly, for inside the wire ($r < R_1$), the total charge increases cubically as radius r increases. Therefore,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{inc}}}{\epsilon_0} \implies E(r)(2\pi rH) = \frac{\rho(\pi r^2 H)}{\epsilon_0} \implies \boxed{E(r) = \frac{\rho r}{2\epsilon_0}}.$$

For the region between the conductor and the shell ($R_1 < r < R_2$), the total charge is no longer dependent on the radius r , hence,

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{\text{inc}}}{\epsilon_0} \implies E(r)(2\pi rH) = \frac{\rho(\pi R_1^2 H)}{\epsilon_0} \implies \boxed{E(r) = \frac{\rho R_1^2}{2\epsilon_0 r}}.$$

Lastly, in the region outside the cable, given that the net charge is both zero, this makes the total charge inclosed $r > R_2$ equal to zero. Therefore,

$$\boxed{E(r) = 0.}$$

□